Bastion / SSH Jump Host Design

Purpose

The pfSense firewall doubles as the **SSH bastion host** for the Virtual SOC environment. This ensures: - No internal systems are directly exposed to ext_net. - All administrative access flows through a single hardened gateway. - Analysts have a secure, auditable entrypoint. - Attack paths remain realistic and controlled.

This design supports multi-hop SSH access into internal hosts using pfSense as the ProxyJump node.

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1. Design Requirements

The bastion must: - Allow SSH key–based authentication only. - Allow agent forwarding for nested authentication. - Restrict direct SSH to internal hosts from ext_net. - Operate on pfSense WAN interface. - Work seamlessly with WireGuard or without it.

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2. Enable SSH on pfSense

Navigate:

```
System → Advanced → Admin Access
```

Enable: - ✓ Secure Shell Server - ✓ Enable SSH Agent Forwarding

SSH now listens on pfSense WAN (default port 22).

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3. Firewall Configuration

WAN Rule — Allow SSH to pfSense

```
Firewall → Rules → WAN → Add
Protocol: TCP
Destination: pfSense WAN address
Port: 22
Action: Pass
```

LAN/Internal Hosts

No direct inbound rules from ext_net—must go through pfSense.

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4. User and Key Management

Navigate:

```
System → User Manager → Users
```

For each administrator: 1. Create a pfSense user (non-admin recommended). 2. Paste SSH public key into “Authorized Keys.”

This is the only authentication method allowed.

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5. SSH Client Configuration

Local workstation `~/.ssh/config`:

```
Host pfsense
  HostName <floating-ip or WG endpoint>
  User <pfsense-user>
  IdentityFile ~/.ssh/id_ed25519
  IdentitiesOnly yes

Host internal
  HostName 105.45.5.72
  User ubuntu
  ProxyJump pfsense
  IdentityFile ~/.ssh/id_ed25519
  IdentitiesOnly yes
```

Usage:

```
ssh internal
```

Traffic path:

```
ext_client → pfSense (SSH) → internal host
```

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6. ProxyJump Explanation

ProxyJump forwards traffic transparently: - The local SSH agent authenticates to pfSense. - pfSense forwards the session into LAN. - The internal host sees login from pfSense, not ext_net.

This preserves isolation while enabling administrative workflows.

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7. Agent Forwarding

Agent forwarding allows: - Reuse of local SSH credentials for deeper hops. - No private keys stored on pfSense.

Verify:

```
ssh-add -L
```

Should show keys when connected to pfSense.

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8. WireGuard Integration

When WireGuard is active, the bastion can be reached through the tunnel:

```
ssh pfsense-wg
```

Which uses: - WG IP: 10.5.5.1 - Same user accounts - Same authorized_keys setup

This method avoids needing a floating IP for pfSense.

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9. Security Recommendations

- Disable password authentication entirely.
- Restrict SSH to WireGuard subnet if possible.
- Create non-admin maintenance accounts.

- Rotate keys periodically.
- Log administrative actions through Wazuh.

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10. Summary of Responsibilities

The Bastion / SSH Jump Host provides: - Secure remote administrative access - Enforced multi-hop internal access - A single audit point for SOC operations - Compatibility with both external and VPN-based workflows

pfSense serves as the hardened gatekeeper ensuring the internal SOC environment remains isolated and controlled during all user interactions.